

Differential Dynamic Programming Formulations for Multi-Period Optimal Power Flow

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I. PROBLEM FORMULATIONS

A. Copper-Plate MPOPF

A storage device and grid connection serve demand p_L^t over $\mathcal{T} = \{1, \dots, T\}$. Here $P_B^t > 0$ discharges, and B^t is the end-of-period energy; hence B^0 is initial and B^T is terminal. The sign convention follows [1].

$$\begin{aligned}
 \min_{P_{\text{Subs}}, P_B, B} \quad & \sum_{t=1}^T \left[c^t P_{\text{Subs}}^t + C_B (P_B^t)^2 \right] \Delta t \\
 & + w (B^T - B_{\text{init}})^2 \quad (1a) \\
 \text{s.t.} \quad & P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \quad t \in \mathcal{T} \quad (1b) \\
 & B^0 = B_{\text{init}} \quad (1c) \\
 & B^t = B^{t-1} - P_B^t \Delta t, \quad t \in \mathcal{T} \quad (1d) \\
 & P_{\text{Subs}}^t \geq 0, \quad t \in \mathcal{T} \quad (1e) \\
 & \underline{P}_B \leq P_B^t \leq \overline{P}_B, \quad t \in \mathcal{T} \quad (1f) \\
 & \underline{B} \leq B^t \leq \overline{B}, \quad t \in \mathcal{T} \quad (1g)
 \end{aligned}$$

The base instance is (1a)–(1d), with all three bounds dropped.

B. SOCP MPOPF

Let $\mathcal{N}$, $\mathcal{L}$, $\mathcal{B}$, and $\mathcal{D}$ denote buses, directed branches, battery buses, and PV buses; 0 is the substation and $\mathcal{C}(j)$ contains the children of j . The network MPOPF is

$$\begin{aligned}
 \min \quad & \sum_{t \in \mathcal{T}} c^t S_{\text{base}} P_{\text{Subs}}^t \Delta t \\
 & + \sum_{t \in \mathcal{T}} \sum_{j \in \mathcal{B}} C_B (S_{\text{base}} P_{B,j}^t)^2 \Delta t \\
 & + w \sum_{j \in \mathcal{B}} (B_j^T - B_{\text{init},j})^2 \quad (2) \\
 \text{s.t.} \quad & P_{\text{Subs}}^t - \sum_{(0,k) \in \mathcal{L}} P_{0k}^t = 0, \quad (3) \\
 & P_{\text{Subs}}^t \geq 0, \quad (4) \\
 & Q_{\text{Subs}}^t - \sum_{(0,k) \in \mathcal{L}} Q_{0k}^t = 0, \quad (5) \\
 & \sum_{k \in \mathcal{C}(j)} P_{jk}^t - P_{ij}^t + r_{ij} \ell_{ij}^t = P_{B,j}^t + p_{D,j}^t - p_{L,j}^t, \quad (6) \\
 & \sum_{k \in \mathcal{C}(j)} Q_{jk}^t - Q_{ij}^t + x_{ij} \ell_{ij}^t = q_{D,j}^t - q_{L,j}^t, \quad (7) \\
 & v_j^t = v_i^t - 2(r_{ij} P_{ij}^t + x_{ij} Q_{ij}^t) + (r_{ij}^2 + x_{ij}^2) \ell_{ij}^t, \quad (8) \\
 & (P_{ij}^t)^2 + (Q_{ij}^t)^2 \leq v_i^t \ell_{ij}^t, \quad (9) \\
 & \ell_{ij}^t \geq 0, \quad (10) \\
 & v_0^t = 1.05^2, \quad (11) \\
 & \underline{V}_j \leq v_j^t \leq \overline{V}_j, \quad (12) \\
 & -q_{D,j}^{\max,t} \leq q_{D,j}^t \leq q_{D,j}^{\max,t}, \quad (13) \\
 & q_{D,j}^{\max,t} = \sqrt{S_{D,j}^2 - (p_{D,j}^t)^2}, \quad (14) \\
 & B_j^t = B_j^{t-1} - P_{B,j}^t \Delta t, \quad (15) \\
 & B_j^0 = B_{\text{init},j}, \quad (16) \\
 & \text{soc}_j^{\min} B_{R,j} \leq B_j^t \leq \text{soc}_j^{\max} B_{R,j}, \quad (17) \\
 & -P_{B,j}^R \leq P_{B,j}^t \leq P_{B,j}^R. \quad (18)
 \end{aligned}$$

II. SIMULATION SETUP

TABLE I
COPPER-PLATE INSTANCE DATA

Interval t	1	2	3	4	5	6
Demand p_L^t	1.657	1.888	1.998	1.834	1.623	1.657
Energy price c^t	0.140	0.197	0.175	0.105	0.083	0.140

Period length Δt	1	Initial energy B_{init}	2.0
Throughput C_B	0.05	Terminal weight w	5.0

<i>Bounds, constrained variants only</i>	
$\underline{P}_B, \overline{P}_B$	−0.45, 0.45
$\underline{B}, \overline{B}$	1.20, 1.95

All quantities are in per unit on a 1000-kVA base, 1 p.u. = 1000 kW and 1 p.u.h = 1000 kWh. Bounds apply only to the constrained variants.

III. IMPLEMENTATION

FilterDDP.jl was modified in two ways. First, its optimal-control interface accepts per-stage objective and constraint vectors, representing time-varying data directly without an auxiliary interpolating state. Second, the feeder path replaces large compile-time `SVector`/`SMatrix` workspaces with dynamic, LAPACK-backed factorizations and analytic derivatives.¹ `StaticArrays` remains for compact stage inputs. Tests use Julia 1.12.6, FilterDDP 0.6.0 (revision 513a104), `StaticArrays` 1.9.18, JuMP 1.31.1, and Ipopt 1.15.0.

A. IEEE123C Evaluation

The network model in Section I-B was evaluated on the single-phase IEEE123C feeder (128 buses, 127 branches, 51 batteries, and 51 PV units) for $T \in \{3, 6, 12, 24, 48, 96\}$. Both FilterDDP and the independent centralized JuMP/Ipopt reference include the $r_{ij} \ell_{ij}^t$ and $x_{ij} \ell_{ij}^t$ line-loss terms in the nodal balances. Every centralized solution passed the constraint validator, and FilterDDP returned normally at all six horizons with maximum equality residual below 9×10^{-8} .

At each stage, $n_x = 51$, $n_u = 791$, and $n_c = 562$. FilterDDP therefore forms a 51×51 value Hessian and a 791×791 control Hessian, factors a 562×562 equality block, and factors a 229×229 reduced Hessian. These dimensions are independent of T ; the number of stage factorizations scales with T .

¹The superseded $T = 2$ stall in static solver construction motivated this rewrite. Xu *et al.* similarly use LAPACK for larger problems [2].

TABLE II
IEEE123C SOLUTION TIMES

T	FilterDDP iterations	FilterDDP (s)	Ipopt (s)
24	97	610.77	3.06
96	121	2021.41	12.57

The complete horizon sweep is archived with the reproducibility results.

APPENDIX A FIRST-ORDER DDP DERIVATION

This appendix states the first-order Differential Dynamic Programming workflow directly in MPOPF notation. Equality multipliers are denoted by λ and inequality multipliers by μ . Thus the dynamics multiplier called $\mu[t]$ in the earlier derivation is written λ_{soc}^t here.

A. Centralized Lagrangian

Write every inequality as $g \leq 0$. The Lagrangian of (1) is

$$\begin{aligned} \mathcal{L} = & \sum_{t=1}^T [c^t P_{\text{Subs}}^t + C_B (P_B^t)^2] \Delta t + w(B^T - B_{\text{init}})^2 \\ & + \sum_{t=1}^T \lambda_{\text{bal}}^t (P_{\text{Subs}}^t + P_B^t - p_L^t) \\ & + \sum_{t=1}^T \lambda_{\text{soc}}^t (B^t - B^{t-1} + \Delta t P_B^t) \\ & + \sum_{t=1}^T \mu_{\underline{P}_{\text{Subs}}}^t (-P_{\text{Subs}}^t) \\ & + \sum_{t=1}^T [\mu_{\underline{P}_B}^t (\underline{P}_B - P_B^t) + \mu_{\overline{P}_B}^t (P_B^t - \overline{P}_B)] \\ & + \sum_{t=1}^T [\mu_{\underline{B}}^t (\underline{B} - B^t) + \mu_{\overline{B}}^t (B^t - \overline{B})]. \end{aligned} \quad (19)$$

Four optimality conditions. Here $B^0 = B_{\text{init}}$ is fixed. In addition to primal feasibility, dual feasibility $\mu \geq 0$, and complementary slackness, stationarity gives four equations. The first two hold for $t = 1, \dots, T$, the third for $t = 1, \dots, T-1$, and the fourth at T :

$$0 = c^t \Delta t + \lambda_{\text{bal}}^t - \mu_{\underline{P}_{\text{Subs}}}^t, \quad (20)$$

$$0 = 2C_B P_B^t \Delta t + \lambda_{\text{bal}}^t + \Delta t \lambda_{\text{soc}}^t - \mu_{\underline{P}_B}^t + \mu_{\overline{P}_B}^t, \quad (21)$$

$$0 = \lambda_{\text{soc}}^t - \lambda_{\text{soc}}^{t+1} - \mu_{\underline{B}}^t + \mu_{\overline{B}}^t, \quad (22)$$

$$0 = 2w(B^T - B_{\text{init}}) + \lambda_{\text{soc}}^T - \mu_{\underline{B}}^T + \mu_{\overline{B}}^T. \quad (23)$$

Equations (20) and (21) are the power stationarity conditions visible inside a period. Equations (22) and (23) make those individually optimal periods multi-period optimal by chaining the marginal value of energy through time and terminating that chain correctly. All four residuals were checked against the centralized JuMP/Ipopt solution; the largest was 2.2×10^{-15} for $T \in \{3, 6\}$.

B. Per-Period Decomposition

At sweep k , period t receives $B^{k,t-1}$ from the period just solved and the successor multiplier $\lambda_{\text{soc}}^{k-1,t+1}$ from the previous sweep. It solves

$$\min_{P_{\text{Subs}}^t, P_B^t, B^t} c^t P_{\text{Subs}}^t \Delta t + C_B (P_B^t)^2 \Delta t - \lambda_{\text{soc}}^{k-1,t+1} B^t \quad (24)$$

$$\text{s.t. } P_{\text{Subs}}^t + P_B^t = p_L^t, \quad (25)$$

$$B^t - B^{k,t-1} + \Delta t P_B^t = 0, \quad (26)$$

$$\begin{aligned} P_{\text{Subs}}^t &\geq 0, \quad \underline{P}_B \leq P_B^t \leq \overline{P}_B, \\ \underline{B} &\leq B^t \leq \overline{B}. \end{aligned} \quad (27)$$

For $t = T$, the last term of (24) is replaced by $w(B^T - B_{\text{init}})^2$. After solving, the equality multiplier is read from the local SOC equation,

$$\lambda_{\text{soc}}^{k,t} = -\text{dual}(B^t - B^{k,t-1} + \Delta t P_B^t = 0). \quad (28)$$

The term $-\lambda_{\text{soc}}^{k-1,t+1} B^t$ inserts the successor term missing from a standalone OPF's stationarity in B^t . The earlier program wrote the equivalent expression

$$\lambda_{\text{soc}}^{k-1,t+1} (B^{k-1,t+1} - B^t + \Delta t P_B^{t+1}). \quad (29)$$

Because $B^{k-1,t+1}$ is fixed and the auxiliary P_B^{t+1} is not the next period's actual decision, only the $-\lambda_{\text{soc}}^{k-1,t+1} B^t$ part supplies the intended inter-period sensitivity.

C. Sweep

The workflow is

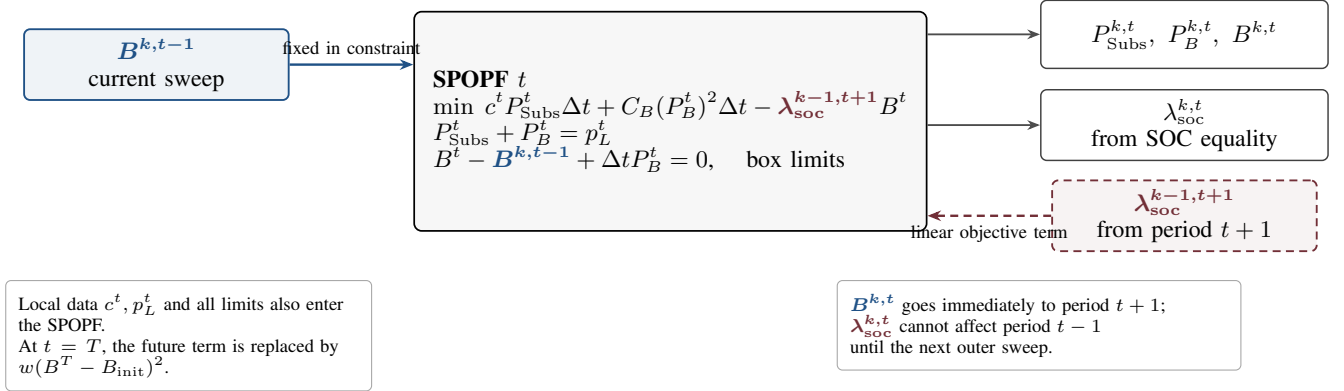
$$t = 1 \rightarrow T : B^{k,t-1} \rightarrow (24) \rightarrow (B^{k,t}, \lambda_{\text{soc}}^{k,t}), \quad k \leftarrow k+1, \quad (30)$$

and stops when

$$\max(\|B^k - B^{k-1}\|_2, \|\lambda_{\text{soc}}^k - \lambda_{\text{soc}}^{k-1}\|_2) < \epsilon. \quad (31)$$

This passes only the first derivative of the future cost. Filter-DDP also passes its curvature V_{BB}^t ; the first-order workflow has no corresponding quantity.

What period t actually receives and solves



Consequence for a $T = 3$ horizon

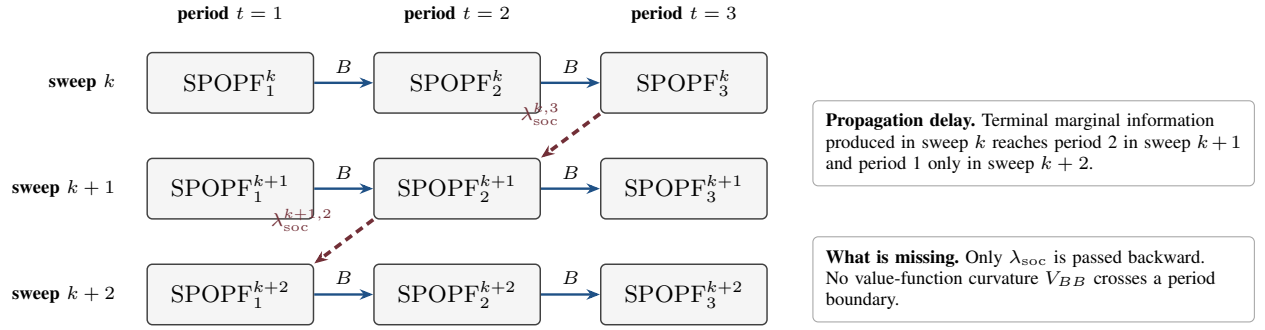


Fig. 1. Information flow in the first-order DDP sweep.

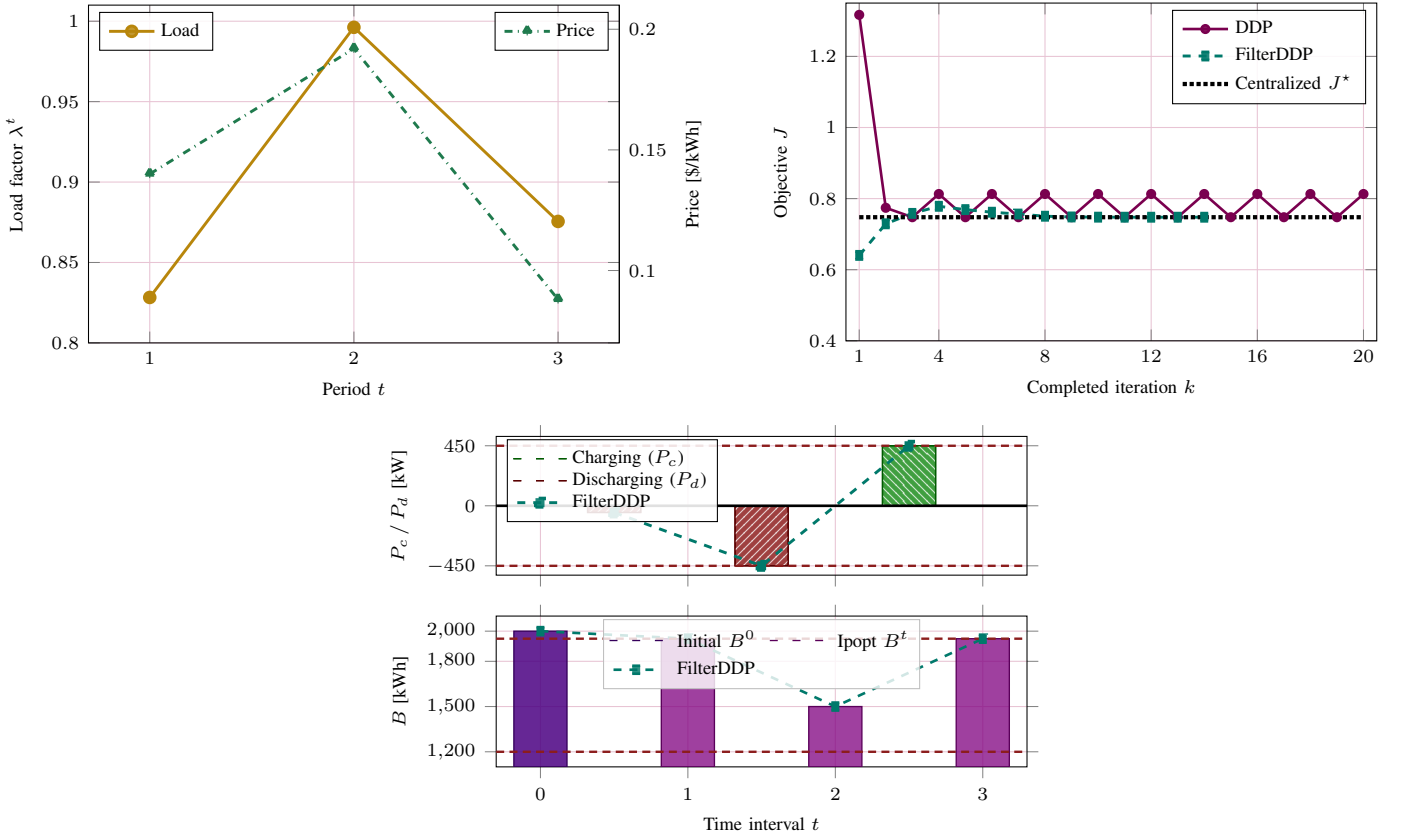


Fig. 2. Three-period inputs, objective trajectories, and optimal battery actions.

TABLE III
SOC EQUALITY MULTIPLIERS DURING THE TWO-CYCLE

k	$\lambda_{\text{soc}}^{k,1}$	$\lambda_{\text{soc}}^{k,2}$	$\lambda_{\text{soc}}^{k,3}$
15	0.135 00	0.088 04	0.314 29
16	0.088 04	0.231 96	0.088 04
17	0.135 00	0.088 04	0.314 29
18	0.088 04	0.231 96	0.088 04
19	0.135 00	0.088 04	0.314 29
20	0.088 04	0.231 96	0.088 04

TABLE IV
IDENTICAL-INSTANCE SOLVER COMPARISON

Method	Outcome	Iterations	Objective J
Centralized Ipopt	optimal	—	0.747650349
FilterDDP	optimal	14	0.747650349
First-order DDP	two-cycle	> 2000	0.747650353/0.812950270

The undamped sweep does not converge. Through 2000 sweeps it remains in an exact period-two cycle: the residual in (31) is 0.602079, and the objective alternates between 0.74765035 and 0.81295027. The centralized objective is 0.747650349; on the other branch the battery dispatch differs by as much as 0.850 p.u. (850 kW). Thus even this three-period copper-plate problem exposes the instability of the first-order iteration. On the identical data, bounds, and initialization, FilterDDP terminates normally in 14 iterations. Its objective differs from centralized Ipopt by 1.48×10^{-10} , while its maximum battery-power and energy differences are 6.50×10^{-9} p.u. and 3.71×10^{-9} p.u.h, respectively. Consequently, the two-cycle is a property of the first-order DDP iteration, not of an unusual or ill-posed optimization instance.

D. LinDistFlow Diagnostic Model

When a full branch-flow instance fails the centralized feasibility gate, the following LinDistFlow equations are used only to diagnose or initialize the network model:

$$\sum_{k \in \mathcal{C}(j)} P_{jk}^t - P_{ij}^t = P_{B,j}^t + p_{D,j}^t - p_{L,j}^t, \quad (32)$$

$$\sum_{k \in \mathcal{C}(j)} Q_{jk}^t - Q_{ij}^t = q_{D,j}^t - q_{L,j}^t, \quad (33)$$

$$v_j^t = v_i^t - 2(r_{ij}P_{ij}^t + x_{ij}Q_{ij}^t). \quad (34)$$

They omit the $r_{ij}\ell_{ij}^t$ and $x_{ij}\ell_{ij}^t$ loss terms, the quadratic current term in the voltage equation, and the SOC current constraint. All device, storage, substation, and voltage-bound constraints are otherwise unchanged. This approximation is not substituted for the loss-aware SOCP in reported OPF comparisons; it is identified explicitly whenever used as a feasibility diagnostic or starting-point construction.

REFERENCES

- [1] A. R. Jha, S. Paul, and A. Dubey, "Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks," in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCCE)*. IEEE, 2025, pp. 04–06.
- [2] M. Xu, S. Gould, and I. Shames, "Line-search filter differential dynamic programming for optimal control with nonlinear equality constraints," 2026.